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Induction Retrieval

Show by induction that for $n \geq 1$

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

We start with a base case, choosing $n=1$

$$\sum_{i=1}^{1} 1^3 = \left[\frac{1(1+1)}{2}\right]^2$$

$$1 = \left[\frac{2}{2}\right]^2$$

$$1 = 1 \quad \checkmark$$

Next we will show that for any valid k, $k+1$ holds. We assume $n=k$ holds

$$\sum_{i=1}^{k} i^3 = \left[\frac{k(k+1)}{2}\right]^2$$

Now we attempt to show $k+1$ also holds

$$\sum_{i=1}^{k+1} i^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

Separate summation

$$\sum_{i=1}^{k} i^3 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$